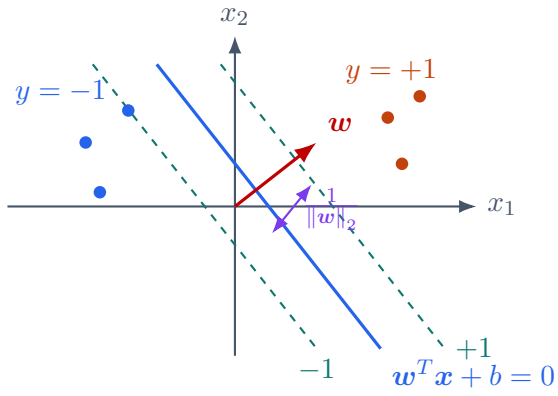
MACHINE LEARNING • MATHEMATICAL NOTES

Geometry of a Linear Classifier

Maximum-Margin Classifier and Hard-Margin SVM



Central idea: Find a separating hyperplane whose nearest training examples are as far away as possible.

Step-by-step intuition, complete mathematical derivations, diagrams, worked examples, convexity proof, and exam revision sheet

Prepared as a self-contained study document

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1 What is a linear classifier?

Suppose each example has d features, represented by a column vector

$$\boldsymbol{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_d \end{bmatrix}^T \in \mathbb{R}^d.$$

A linear classifier separates the feature space using a line (in two dimensions), a plane (in three dimensions), or a *hyperplane* (in d dimensions). Its decision boundary is

Decision boundary

$$\boxed{\boldsymbol{w}^T\boldsymbol{x} + b = 0}$$

Here $\boldsymbol{w} \in \mathbb{R}^d$ is the weight/normal vector and $b \in \mathbb{R}$ is the bias (intercept).

The predicted class for labels in $\{-1, +1\}$ is

$$\boxed{\hat{y} = \text{sign}(\boldsymbol{w}^T\boldsymbol{x} + b).}$$

A positive score predicts $+1$, a negative score predicts -1, and a zero score places the point on the boundary.

1.1 From the ordinary line equation to vector form

Start with the two-dimensional line

$$x_2 = mx_1 + c.$$

Move every term to one side:

$$mx_1 - x_2 + c = 0.$$

With $\boldsymbol{x} = (x_1, x_2)^T$, this becomes

$$\begin{bmatrix} m & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + c = 0.$$

Therefore $\boldsymbol{w} = (m, -1)^T$ and $b = c$. This shows that the general hyperplane equation includes the usual line equation as a special case.

1.2 Why is the weight vector perpendicular to the boundary?

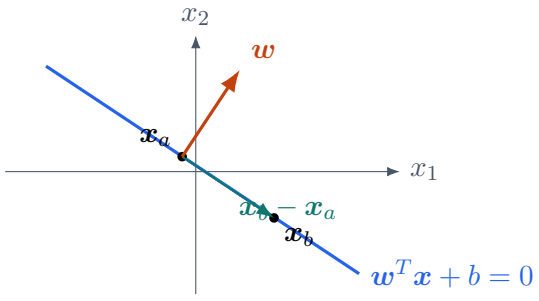
Take any two points $\boldsymbol{x}_a, \boldsymbol{x}_b$ on the boundary. Since both satisfy the boundary equation,

$$\boldsymbol{w}^T\boldsymbol{x}_a + b = 0, \qquad \boldsymbol{w}^T\boldsymbol{x}_b + b = 0.$$

Subtracting gives

$$\boldsymbol{w}^T(\boldsymbol{x}_a - \boldsymbol{x}_b) = 0.$$

The vector $\boldsymbol{x}_a - \boldsymbol{x}_b$ points along the boundary. Its dot product with $\boldsymbol{w}$ is zero, so $\boldsymbol{w}$ is perpendicular to the boundary. The vector $\boldsymbol{w}$ is called the **normal vector**.



2 Score, probability, and distance

Define the raw decision score

$$f(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{x} + b.$$

Its sign determines the predicted class. However, the score magnitude is not itself a physical distance: multiplying both $\boldsymbol{w}$ and b by a positive constant multiplies the score but leaves the boundary unchanged.

2.1 Connection to logistic regression

The sigmoid function is

$$\sigma(z) = \frac{1}{1+e^{-z}}.$$

Logistic regression models $P(y = +1 \mid \boldsymbol{x}) = \sigma(\boldsymbol{w}^T\boldsymbol{x} + b)$. Because σ is increasing, a score of zero corresponds to probability 0.5. Logistic regression estimates probabilities; a maximum-margin classifier instead chooses a separating boundary by maximizing its geometric margin. The decision-boundary form can be the same even though the training objectives differ.

3 Deriving point-to-hyperplane distance

Let the hyperplane be $H : \boldsymbol{w}^T\boldsymbol{x} + b = 0$ and let $\boldsymbol{p}$ be the closest point on H to a point $\boldsymbol{x}$. The shortest segment from $\boldsymbol{p}$ to $\boldsymbol{x}$ is perpendicular to H, and hence parallel to $\boldsymbol{w}$. For some scalar t,

$$\boldsymbol{x} - \boldsymbol{p} = t\boldsymbol{w}.$$

Take the dot product with $\boldsymbol{w}$:

$$\boldsymbol{w}^T(\boldsymbol{x} - \boldsymbol{p}) = t\,\|\boldsymbol{w}\|_2^2\,.$$

Because $\boldsymbol{p}$ lies on the hyperplane, $\boldsymbol{w}^T\boldsymbol{p} = -b$. Hence

$$t = \frac{\boldsymbol{w}^T\boldsymbol{x} + b}{\|\boldsymbol{w}\|_2^2}.$$

The Euclidean distance is $\|\boldsymbol{x} - \boldsymbol{p}\|_2 = |t|\,\|\boldsymbol{w}\|_2$, giving the key formula:

Point-to-hyperplane distance

$$\boxed{d(\boldsymbol{x}, H) = \frac{|\boldsymbol{w}^T\boldsymbol{x} + b|}{\|\boldsymbol{w}\|_2}} \qquad \boxed{\delta(\boldsymbol{x}, H) = \frac{\boldsymbol{w}^T\boldsymbol{x} + b}{\|\boldsymbol{w}\|_2}}$$

The first is the nonnegative distance; the second is the signed distance.

3.1 Numerical example

Consider $2x_1 + 2x_2 - 6 = 0$. Here $\boldsymbol{w} = (2, 2)^T$, $b = -6$, and $\|\boldsymbol{w}\|_2 = 2\sqrt{2}$. At $\boldsymbol{x} = (1, 1)^T$, the score is $2 + 2 - 6 = -2$. Thus

$$\delta = \frac{-2}{2\sqrt{2}} = -\frac{1}{\sqrt{2}}, \qquad d = \frac{1}{\sqrt{2}} \approx 0.707.$$

The negative sign tells us which side of the boundary contains the point; the distance itself is nonnegative.

3.2 Why divide by the norm? Scale invariance

For any $c > 0$, define $\boldsymbol{w}' = c\boldsymbol{w}$ and $b' = cb$. Then

$$(\boldsymbol{w}')^T\boldsymbol{x} + b' = c(\boldsymbol{w}^T\boldsymbol{x} + b) = 0$$

defines exactly the same hyperplane. Also $\|\boldsymbol{w}'\|_2 = c\,\|\boldsymbol{w}\|_2$. Therefore

$$\frac{|(\boldsymbol{w}')^T\boldsymbol{x} + b'|}{\|\boldsymbol{w}'\|_2} = \frac{c|\boldsymbol{w}^T\boldsymbol{x} + b|}{c\,\|\boldsymbol{w}\|_2} = \frac{|\boldsymbol{w}^T\boldsymbol{x} + b|}{\|\boldsymbol{w}\|_2}.$$

Geometric interpretation

The equation used to describe a line can change scale, but the actual line and physical distances do not. Dividing by $\|\boldsymbol{w}\|_2$ removes the arbitrary scale.

4 Functional margin and geometric margin

Assume the training data are $\mathcal{D} = \{(\boldsymbol{x}^{(i)}, y^{(i)})\}_{i=1}^N$, with $y^{(i)} \in \{-1, +1\}$.

4.1 Functional margin

For one training example, define

$$\boxed{\widehat{\gamma}^{(i)} = y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b).}$$

Multiplying by the label makes the sign consistent across the two classes:

- If $y^{(i)} = +1$, a correct prediction requires $\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b > 0$.
- If $y^{(i)} = -1$, a correct prediction requires $\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b < 0$.

Thus a correctly classified point has positive functional margin, a misclassified point has negative functional margin, and a point on the boundary has zero margin. Its magnitude changes if $(\boldsymbol{w}, b)$ is rescaled.

4.2 Geometric margin

Divide the functional margin by the norm of the normal vector:

$$\boxed{\gamma^{(i)} = \frac{y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b)}{\|\boldsymbol{w}\|_2}.}$$

This is the signed perpendicular distance, oriented toward the correct class. The functional and geometric margins have the same sign, but only the geometric margin is invariant to positive rescaling.

	Functional margin	Geometric margin
Formula	$y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b)$	$\dfrac{y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b)}{\|\boldsymbol{w}\|_2}$
Scale invariant?	No	Yes, for positive rescaling
Interpretation	Algebraic score	Signed perpendicular distance
Use	Convenient constraints	Actual geometry

4.3 A scale example

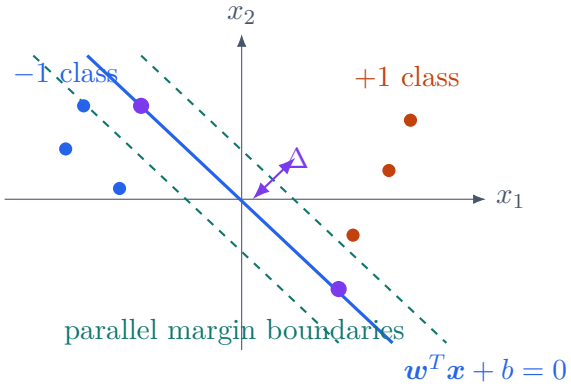
Let $\boldsymbol{w} = (3, 4)^T$, $b = -5$, and $\boldsymbol{x} = (2, 1)^T$ with label $+1$. Then the score is $3(2) + 4(1) - 5 = 5$, and $\|\boldsymbol{w}\|_2 = 5$. Functional margin $= 5$; geometric margin $= 5/5 = 1$. If $(\boldsymbol{w}, b)$ is doubled, the functional margin becomes 10, but the geometric margin remains $10/10 = 1$.

5 The margin of an entire dataset

The minimum geometric margin is

$$\boxed{\Delta(\boldsymbol{w}, b) = \min_{i=1,\dots,N} \gamma^{(i)} = \min_i \frac{y^{(i)}(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b)}{\|\boldsymbol{w}\|_2}.}$$

For a linearly separable dataset, all signed margins are positive. The minimum is controlled by the closest examples. Maximizing this minimum distance gives the maximum-margin classifier.



The margin Δ is the distance from the central boundary to the nearest point. The full width of the corridor between the two margin boundaries is 2Δ.

6 Why we can normalize the functional margin to one

Suppose a separating hyperplane has positive minimum functional margin

$$m = \min_i y^{(i)}(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b) > 0.$$

Scale the parameters by $c = 1/m$. The new minimum functional margin becomes

$$\min_i y^{(i)}((c\boldsymbol{w})^T \boldsymbol{x}^{(i)} + cb) = cm = 1.$$

This rescaling does not move the hyperplane or change geometric distances. Therefore, without loss of generality, we may use the normalization

Canonical normalization

$$\boxed{\min_i y^{(i)}(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b) = 1}$$

which is equivalent to the constraints

$$\boxed{y^{(i)}(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b) \geq 1, \quad i = 1, \dots, N.}$$

The number 1 is not a physical distance. It is a convenient choice of scale for the algebraic scores. At an optimum, at least one point is tight under the standard nondegenerate hard-margin solution.

6.1 The scale constant c^*

If the original parameters are $(\boldsymbol{w}, b)$, write $\boldsymbol{w}' = c^*\boldsymbol{w}$ and $b' = c^*b$. To make the minimum functional margin equal to one, choose

$$\boxed{c^* = \frac{1}{\min_i y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b)}}$$

provided the original hyperplane correctly separates the data. Under this normalization, the minimum geometric margin is

$$\boxed{\Delta(\boldsymbol{w}', b') = \frac{1}{\|\boldsymbol{w}'\|_2}.}$$

This is the bridge between geometric margin and the norm of the weight vector.

7 Deriving the hard-margin SVM optimization problem

The goal is to maximize the minimum geometric margin:

$$\max_{\boldsymbol{w},b} \min_i \frac{y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b)}{\|\boldsymbol{w}\|_2}.$$

This expression is awkward because it combines a minimum with a norm in the denominator. Apply the canonical normalization from the previous section. The constraints become $y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b) \geq 1$, and the margin is $1/\|\boldsymbol{w}\|_2$.

Therefore the equivalent normalized objective is

$$\max_{\boldsymbol{w},b} \frac{1}{\|\boldsymbol{w}\|_2} \quad \text{subject to} \quad y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b) \geq 1.$$

Since $\|\boldsymbol{w}\|_2 > 0$, maximizing its reciprocal is equivalent to minimizing the norm:

$$\min_{\boldsymbol{w},b} \|\boldsymbol{w}\|_2 .$$

Minimizing a nonnegative norm is equivalent to minimizing its square. Multiplying the objective by a positive constant also does not change the minimizer. We conventionally use a factor of $1/2$ because it simplifies derivatives.

Standard hard-margin SVM primal problem

$$\boxed{\begin{aligned} \min_{\boldsymbol{w},b} \quad & \frac{1}{2}\|\boldsymbol{w}\|_2^2 \\ \text{subject to} \quad & y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b) \geq 1, \\ & i = 1, \ldots, N. \end{aligned}}$$

The factor $1/2$ is chosen because

$$\nabla_{\boldsymbol{w}} \left(\frac{1}{2}\boldsymbol{w}^T\boldsymbol{w}\right) = \boldsymbol{w},$$

whereas the derivative of $\boldsymbol{w}^T\boldsymbol{w}$ is $2\boldsymbol{w}$.

8 The two margin boundaries and their width

The central boundary is $\boldsymbol{w}^T\boldsymbol{x} + b = 0$. Under the normalization, the parallel margin boundaries are

$$\boxed{\boldsymbol{w}^T\boldsymbol{x} + b = +1 \qquad \text{and} \qquad \boldsymbol{w}^T\boldsymbol{x} + b = -1.}$$

The distance from a point on either margin boundary to the central boundary is given by the point-to-hyperplane formula:

$$d_+ = \frac{|1-0|}{\|\boldsymbol{w}\|_2} = \frac{1}{\|\boldsymbol{w}\|_2}, \qquad d_- = \frac{|-1-0|}{\|\boldsymbol{w}\|_2} = \frac{1}{\|\boldsymbol{w}\|_2}.$$

The total width is consequently

$$\boxed{\text{margin width} = d_+ + d_- = \frac{2}{\|\boldsymbol{w}\|_2}.}$$

Thus maximizing the margin is equivalent to minimizing $\|\boldsymbol{w}\|_2$ or $\frac{1}{2}\|\boldsymbol{w}\|_2^2$.

9 Why the optimization problem is convex

A constrained optimization problem is convex when its objective is convex and its feasible region is convex.

9.1 The objective is convex

Expand the objective:

$$J(\boldsymbol{w}) = \frac{1}{2}\|\boldsymbol{w}\|_2^2 = \frac{1}{2}\sum_{j=1}^{d} w_j^2.$$

Its gradient and Hessian are

$$\nabla_{\boldsymbol{w}} J = \boldsymbol{w}, \qquad \boxed{\nabla_{\boldsymbol{w}}^2 J = I_d.}$$

The identity matrix is positive definite, since $\boldsymbol{v}^T I_d \boldsymbol{v} = \|\boldsymbol{v}\|_2^2 > 0$ for any nonzero $\boldsymbol{v}$. Hence the objective is strictly convex in $\boldsymbol{w}$. It is independent of b, so it is not strictly convex in the combined pair $(\boldsymbol{w}, b)$.

9.2 The feasible region is convex

Each constraint can be written

$$y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b) - 1 \geq 0.$$

The left-hand side is affine in $(\boldsymbol{w}, b)$. An affine inequality defines a half-space, and half-spaces are convex. The intersection of all these half-spaces is also convex. Therefore the feasible region is convex.

Convexity conclusion

$$\boxed{\text{Convex objective} + \text{convex feasible set} \Rightarrow \text{convex optimization problem.}}$$

For linearly separable training data, the hard-margin constraints are feasible and the global optimum can be found by convex optimization methods.

10 Worked example: a simple maximum-margin separator

Consider four points:

Point and coordinates	Label
$A = (2,0)^T$	$+1$
$B = (2,2)^T$	$+1$
$C = (0,0)^T$	-1
$D = (0,2)^T$	-1

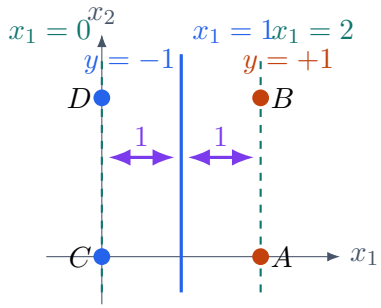
The closest positive points have $x_1 = 2$ and the closest negative points have $x_1 = 0$. The midpoint boundary is $x_1 = 1$. Choose

$$\boldsymbol{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad b = -1.$$

Then the boundary is $x_1 - 1 = 0$. Check the margin constraint for every point:

$$\begin{aligned} A: &\quad (+1)(2-1) = 1, \\ B: &\quad (+1)(2-1) = 1, \\ C: &\quad (-1)(0-1) = 1, \\ D: &\quad (-1)(0-1) = 1. \end{aligned}$$

All four examples satisfy the constraints exactly. Also $\|\boldsymbol{w}\|_2 = 1$, so the one-sided margin is 1 and the total width is 2.



This example illustrates that the decision boundary is halfway between the two closest class boundaries. The support vectors are the points that touch the margin boundaries.

11 Support vectors

Support vectors are training examples that attain equality in the normalized margin constraint:

$$\boxed{y^{(i)}(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b) = 1.}$$

They lie on the margin boundaries and are closest to the central decision boundary. These examples are crucial because they determine the tightest constraints. Points far from the boundary can often move a little without changing the fitted separator, whereas moving a support vector can change the maximum-margin solution.

12 Hard margin versus soft margin

The hard-margin SVM requires perfect linear separability. If classes overlap or no hyperplane satisfies all constraints, the hard-margin problem is infeasible. A soft-margin SVM introduces slack variables $\xi_i \geq 0$:

$$\boxed{\begin{aligned} \min_{\boldsymbol{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i=1}^{N} \xi_i \\ \text{subject to} \quad & y^{(i)}(\boldsymbol{w}^T \boldsymbol{x}^{(i)} + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0, \quad i = 1, \ldots, N. \end{aligned}}$$

The parameter $C > 0$ controls the trade-off between a wide margin and violations of the margin constraints. Large C penalizes violations more heavily; smaller C permits more violations in exchange for stronger regularization.

13 Common mistakes and exam traps

Avoid these mistakes

1. **Maximizing the norm.** The correct objective is to minimize $\frac{1}{2}\|\boldsymbol{w}\|_2^2$, not maximize it.
2. **Forgetting the label.** The constraint must be $y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)}+b)\geq 1$. Without $y^{(i)}$, negative examples would be constrained incorrectly.
3. **Calling 1 the physical margin.** Under normalization, 1 is the functional margin level. The one-sided geometric margin is $1/\|\boldsymbol{w}\|_2$.
4. **Forgetting the factor of two.** The distance from the central boundary to one margin boundary is $1/\|\boldsymbol{w}\|_2$; the full corridor width is $2/\|\boldsymbol{w}\|_2$.
5. **Assuming every dataset is hard-margin separable.** Overlapping classes may make the hard-margin constraints infeasible; use a soft-margin formulation when appropriate.
6. **Thinking rescaling changes the geometry.** Positive scaling changes the equation's numerical coefficients, not the hyperplane or geometric distances.

14 Formula sheet for last-minute revision

Eight formulas to know

$$\begin{aligned}
\text{Decision boundary:}\quad & \boldsymbol{w}^T\boldsymbol{x}+b=0\\
\text{Prediction:}\quad & \hat{y}=\operatorname{sign}(\boldsymbol{w}^T\boldsymbol{x}+b)\\
\text{Point distance:}\quad & d(\boldsymbol{x},H)=\frac{|\boldsymbol{w}^T\boldsymbol{x}+b|}{\|\boldsymbol{w}\|_2}\\
\text{Functional margin:}\quad & \widehat{\gamma}^{(i)}=y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)}+b)\\
\text{Geometric margin:}\quad & \gamma^{(i)}=\frac{y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)}+b)}{\|\boldsymbol{w}\|_2}\\
\text{Dataset margin:}\quad & \Delta=\min_i \gamma^{(i)}\\
\text{One-sided / full width:}\quad & \Delta=\frac{1}{\|\boldsymbol{w}\|_2},\quad W=\frac{2}{\|\boldsymbol{w}\|_2}\\
\text{Hard-margin SVM:}\quad & \min_{\boldsymbol{w},b}\frac{1}{2}\|\boldsymbol{w}\|_2^2\ \text{ s.t. }\ y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)}+b)\geq 1
\end{aligned}$$

15 Self-test questions

Try to answer each question before checking the key.

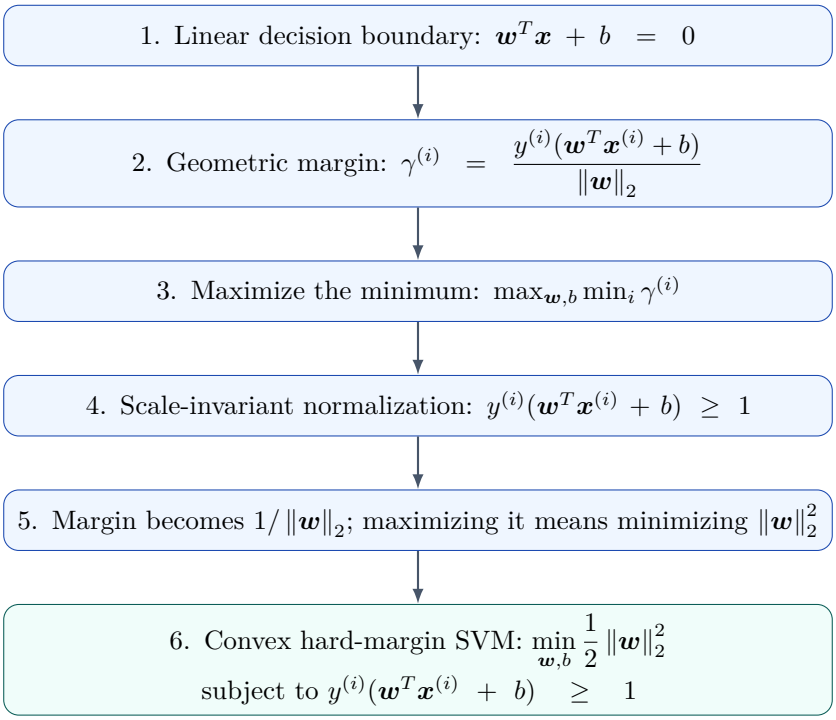
1. What is the perpendicular distance from $\boldsymbol{x}$ to $\boldsymbol{w}^T\boldsymbol{x}+b=0$?
2. Why do we multiply the score by the label $y^{(i)}$ in the SVM constraint?
3. Under the normalized constraints, what is the one-sided geometric margin?
4. Why is the hard-margin SVM a convex optimization problem?
5. What is the full width between the two margin boundaries?

6. What does a support vector satisfy in the canonical formulation?

Answer key

1. $|\boldsymbol{w}^T\boldsymbol{x} + b| / \|\boldsymbol{w}\|_2$.
2. It makes the signed score positive for correctly classified examples of either class.
3. $1/\|\boldsymbol{w}\|_2$.
4. The objective is a convex quadratic and each constraint is affine, giving a convex feasible set.
5. $2/\|\boldsymbol{w}\|_2$.
6. $y^{(i)}(\boldsymbol{w}^T\boldsymbol{x}^{(i)} + b) = 1$ for a point on a margin boundary.

The full derivation in one diagram



Remember the logic

Distance $\longrightarrow$ minimum margin $\longrightarrow$ normalization $\longrightarrow$ minimum norm $\longrightarrow$ convex optimization.